# Qatar-1b

**R-1659**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_250408` · created 2026-08-02T09:09:13+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2460773.9225 +/- 0.0027 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.158 +/- 0.021 |
| Transit depth (Rp/Rs)^2 | 2.5 +/- 0.65 [%] |
| Orbital Inclination (inc) | 84.37 +/- 1.79 |
| Airmass coefficient 1 (a1) | 0.978 +/- 0.021 |
| Airmass coefficient 2 (a2) | 0.012 +/- 0.011 |
| Scatter in the residuals of the lightcurve fit is | 2.11 % |
| Best Comparison Star | #3 - [55, 159] |
| Optimal Method | PSF photometry |
| Transit Duration (day) | 0.07 +/- 0.01 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.158 ± 0.021 vs 0.1463 (deviation 0.56σ) |
| Duration fit vs published | 0.07 d vs 0.0692 d (deviation 0.08σ) |
| Mid-time O–C | 1.8 min |
| Depth SNR | 7.5 |
| Residual scatter | 2.11 % |

## Light curve

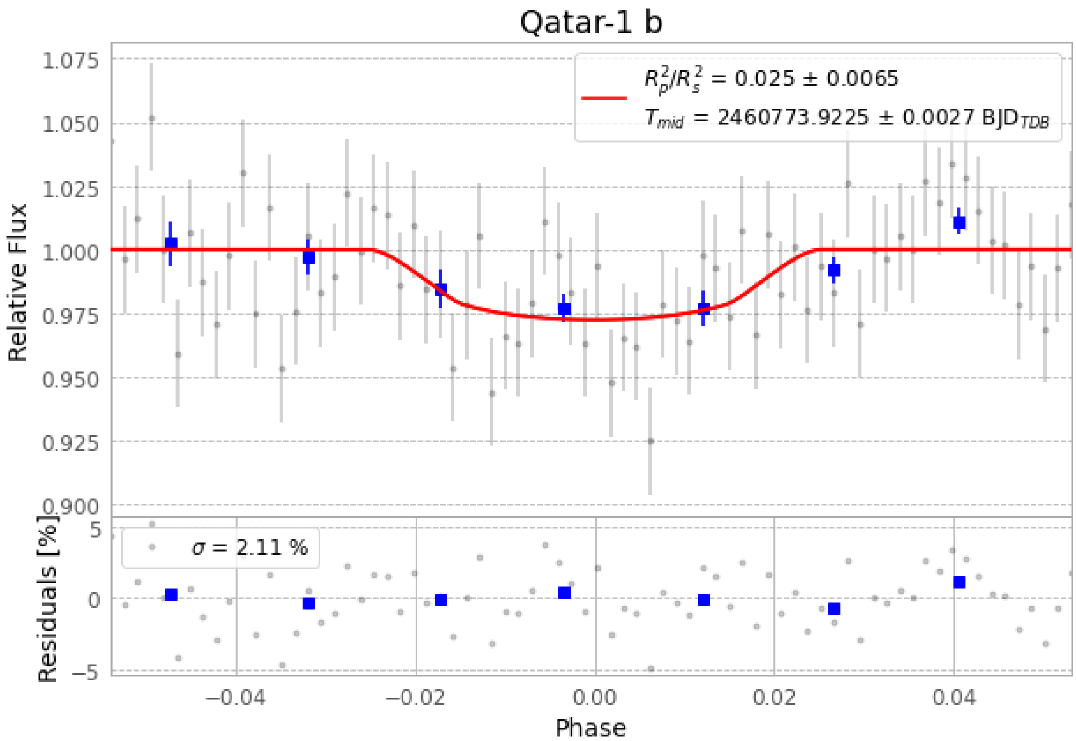


Figure 1 — FinalLightCurve_Qatar-1 b_2025-04-08.png (SHA-256 591e93f5a3064826…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 162], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2186e5d07c66025a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

74 FITS frames (49 MB), `Qatar-1250408081815.FITS` → `Qatar-1250408115715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250408.log (3ff219a74e1a…), FinalLightCurve_Qatar-1 b_2025-04-08.png (591e93f5a306…), FinalLightCurve_Qatar-1 b_2025-04-08.csv (189a54045a13…)

## Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)